GCSE · OCR · Computer Science

52 mins 14 questions

Boolean Logic Diagrams

Boolean Logic Diagrams

Boolean Logic Diagrams / Combining Boolean Operators

Easy (5 questions)	/10
Medium (5 questions)	/26
Hard (4 questions)	/16
Total Marks	**/52**

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Easy Questions

1 (a) Complete the truth table in **Fig. 1** for the Boolean statement P = NOT(A AND B).

A	B	P
0	0	1
0	1	
1	0	
1	1	0

Fig. 1

(2 marks)

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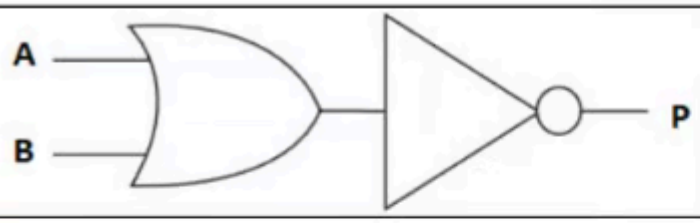
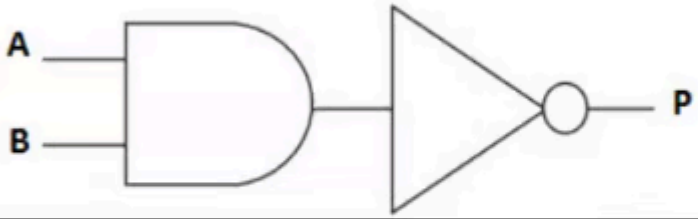
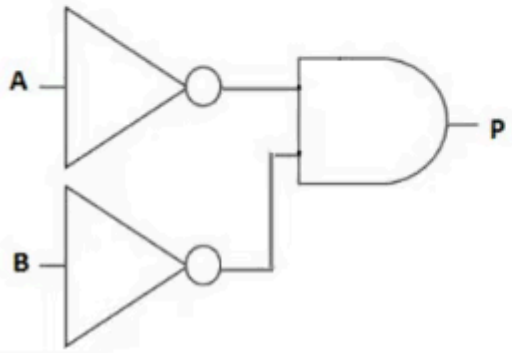
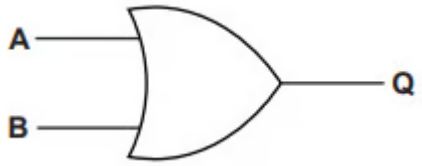
(b) **Tick (✓) one** box to identify the **correct logic diagram** for P = NOT(A AND B).

P = NOT (A AND B)	Tick (✓) one box
	
	
	

(1 mark)

2 **Complete** the truth table for the following logic gate.



A	B	Q
0	0	0
0	1	1
	0	
1		

(4 marks)

3 **State** the **number rows** needed in a truth table for a Boolean expression with four inputs?

..... **(1 mark)**

4 **What is the result** of the Boolean expression A ∨ B when A = 0 and B = 1?

..... **(1 mark)**

5 **Write** the Boolean expression for a logic circuit with an **AND** gate followed by a **NOT** gate.

..... **(1 mark)**

Medium Questions

1 (a) A fast food restaurant offers half-price meals if the customer is a student or has a discount card.
The offer is not valid on Saturdays.

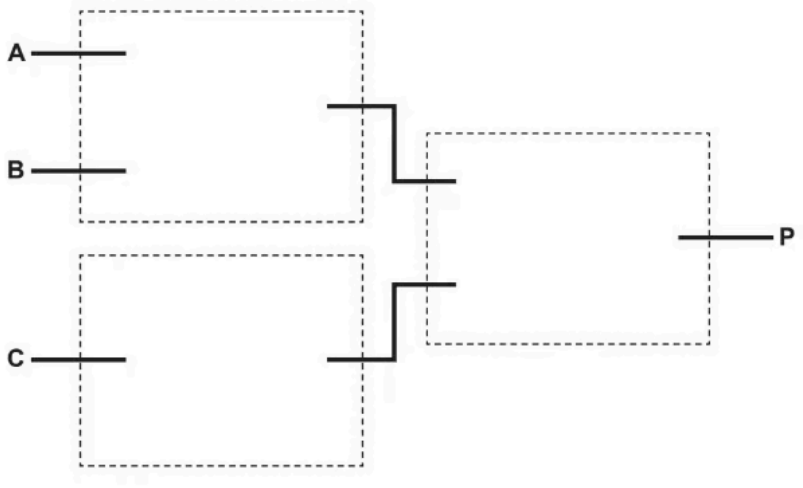
A computer system is used to identify whether the customer can have a half-price meal.

The table identifies the three inputs to the computer system:

Input	Value
A	Is a student
B	Has a discount card
C	The current day is Saturday

The logic system **P = (A OR B) AND NOT C** is used

Complete the following logic diagram for **P = (A OR B) AND NOT C** by drawing one logic gate in each box.



(3 marks)

(b) A **truth table** can be produced for this logic circuit.

Describe the **purpose** of a truth table.

.....

..... **(2 marks)**

(c) **State how many rows** (excluding any headings) would be required in a truth table for the logic expression:

P = (A OR B) AND NOT C

..... **(1 mark)**

2 (a) **Draw** the **logic diagram** for the logic system **P = A OR (B AND C)**

(3 marks)

(b) **Complete** the **truth table** for the logic system **P = NOT (A OR B)**

A	B	P
0	0	1
0	1	-
1	0	-
-	-	-

(4 marks)

3 (a) The logic diagram below (Fig. 2) shows a system made up of two connected logic gates.

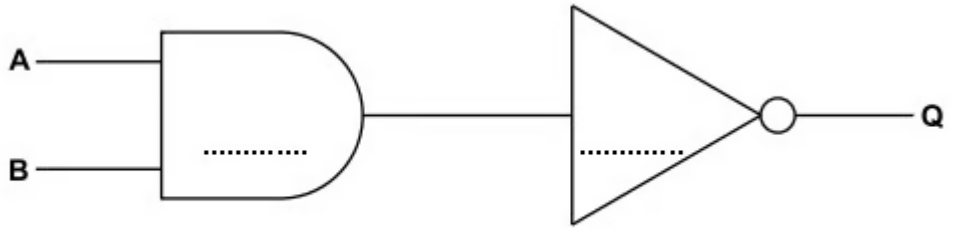


Fig. 2

Label the **names** of the **two gates** on the diagram above

(2 marks)

(b) **Complete** the **truth table** below to show the output from the logic system in **part(a)**.

A	B	C
0	0	
0	1	
1	0	
1	1	

(4 marks)

4 **Draw** the **logic diagram** represented by $Q = A \vee \neg B$

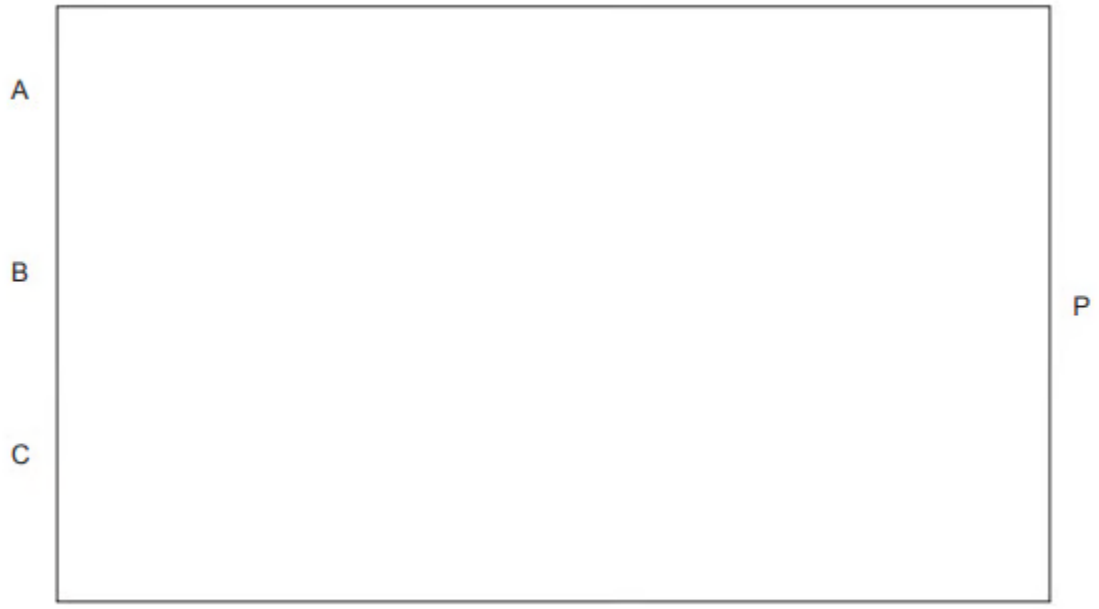
(2 marks)

5 (a) **Complete** the truth table for **P = (A AND B) OR C**

A	B	C	P
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(4 marks)

(b) **Draw** a logic circuit for **P = NOT A AND (B OR C)**



(1 mark)

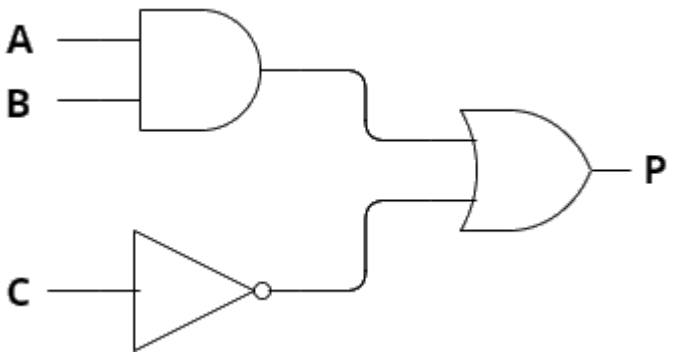
Hard Questions

1 **Draw** the **logic diagram** represented by:

P =(A OR NOT B) AND (A OR C)

(5 marks)

2 **Write** the **Boolean expression** represented by the logic diagram below:



.....

.....

..... **(3 marks)**

3 **Complete** the **truth table** for the Boolean expression:

P = (A AND B) AND NOT C

A	B	C	P
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	-
-	-	-	-
-	-	-	-
-	-	-	-

(5 marks)

4 Given the logic circuit with inputs A, B, and C:

P = (A OR B) AND NOT C

State the **outputs** when A = 1, B = 0, and C = 1.

.....

.....

..... **(3 marks)**